

Lecture 3: Volumes by Slicing; Disks and Washers

Monday, August 31 • Sections 2.2

• Guided Notes •

How to Use These Guided Notes: These guided notes are designed to help you organize the main ideas from each lecture but

1. we will not complete every example in class.
2. Although most definitions and theorems are included, you may need to add additional notes, examples, or explanations during class.
3. Occasionally, you will see sections labeled **Deeper Insight**. These sections provide interesting background information and help develop a deeper understanding of calculus, but they are not part of the core Calculus II content.

Today's lecture is devoted to another application of integration: determining the volumes of three-dimensional objects.

Deeper Insight

Last week, we saw that

Integrating lengths produces areas.

A vertical or horizontal segment has a length. By accumulating infinitely many lengths, we can determine an area.

This week, we extend this idea

Integrating areas produces volumes.

Understanding what quantity is being accumulated is often more important than memorizing a formula.

In this lesson, we will use integration to calculate the volumes of special three-dimensional objects: objects that can be sliced into cross-sections, parallel to one another and perpendicular to one of the coordinate axes, whose areas are easy to determine. By integrating the area of these cross-sections, we can find the volume of the entire solid. This process is called the **slicing method**. A particularly important application is to solids of revolution. For solids of revolution, the slices are circles or washers, and the slicing method is known as the **disk/washer method**.

The key word throughout this lesson is **slice**. A slice represents a cross-section of the solid. In general, it represents a closed planar region such as a square, triangle, rectangle, or ellipse. For solids of revolution, the slices represent circles or washers.

Definition 3.1 – Cross-section of a solid

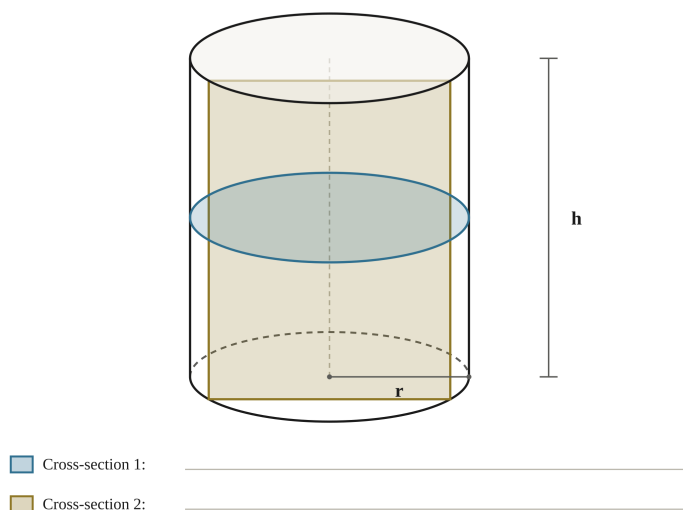
A cross-section of a solid is the planar region formed by intersecting the solid with a plane.

Understanding the definition

Cross-sections of a sphere ^a are _____.

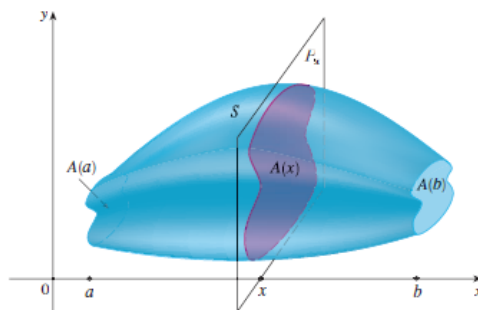
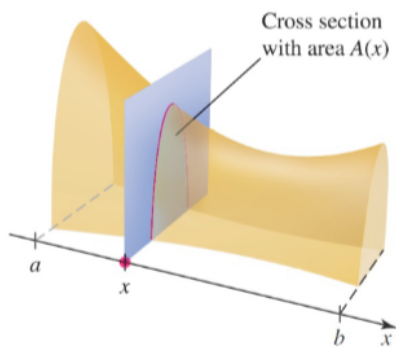
This is a right cylinder with a circular base. What shapes do cross-sections have? How is the corresponding plane intersecting the cylinder, parallel or perpendicular to the base?

Right Circular Cylinder — Two Cross-Sections



^aDesmos files: Click on the wrench in the upper right corner → More Options → Translucent surfaces

Definition 3.2 – Slicing by parallel planes



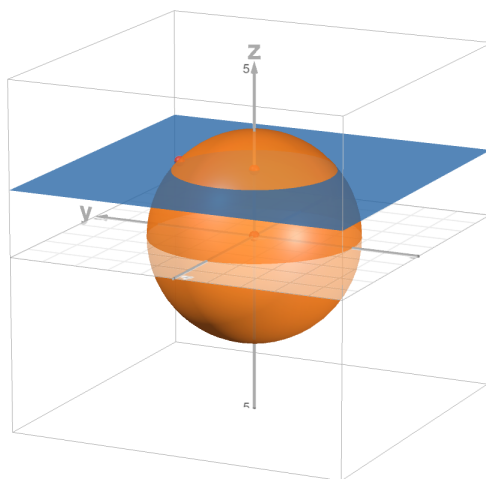
Suppose the three dimensional object is inside the cube $[a, b] \times [c, d] \times [h, k]$.

For every $x \in [a, b]$, let $A(x)$ denote the area of the cross section perpendicular to the x -axis.

If $A(x)$ is a continuous function, then the volume of the solid is _____.

Example 3.1 – Volume of a sphere

Let us determine the volume of a sphere with radius R using the slicing method.



Each cross-section, parallel to the XY -plane, is a circle. The radius of the circle depends on the height of the plane intersecting the sphere. Suppose $A(z)$ is its area. Then the volume of the sphere

$$\int_{\boxed{}}^{\boxed{}} A(z) \, dz \quad \text{or equivalently,} \quad \boxed{} \int_0^{\boxed{}} A(z) \, dz.$$

Let $r(z)$ denote the radius of the circle with center at $(0, 0, z)$, cross-section of the sphere with a plane parallel to the XY -plane, as shown in the figure above. The area of this circle is $A(z) = \pi(r(z))^2$. All we need to do now is

- determine the radius of that circle (function depending on z),
- set up the definite integral equal to the volume of the sphere, and
- evaluate it.

Hopefully, we will get the same formula that Archimedes found more than two thousand years ago ¹.

¹If you want to know a bit about Archimedes work, see the video posted in Canvas, in the first module

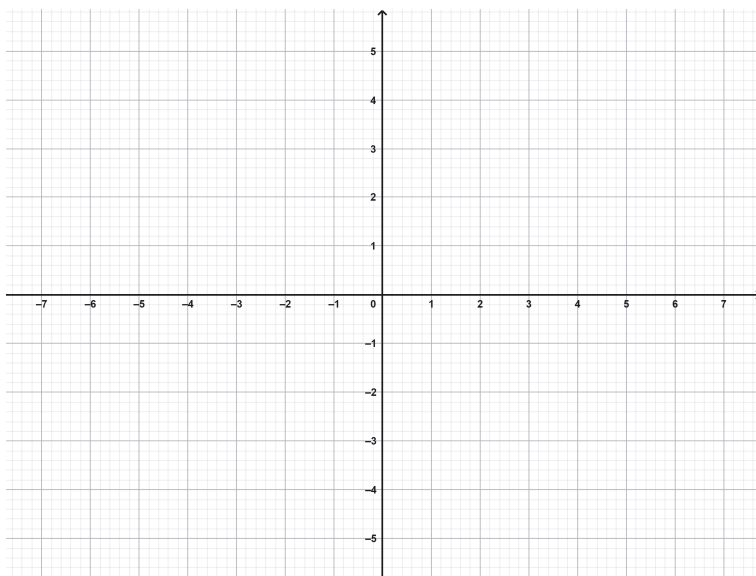
Try it yourself.

The volume of a right cylinder with circular base of radius R and height H is $\pi R^2 H$. Can you get the same formula using the slicing method?

Can you, using the slicing method, prove that the volume of a cone with base of radius R and height H is $\frac{1}{3}$ the volume of the cylinder with the same dimensions?

Example 3.2 – No image of the solid provided

Find the volume of a solid with a circular base of radius 1. Parallel cross-sections perpendicular to the base are equilateral triangles.

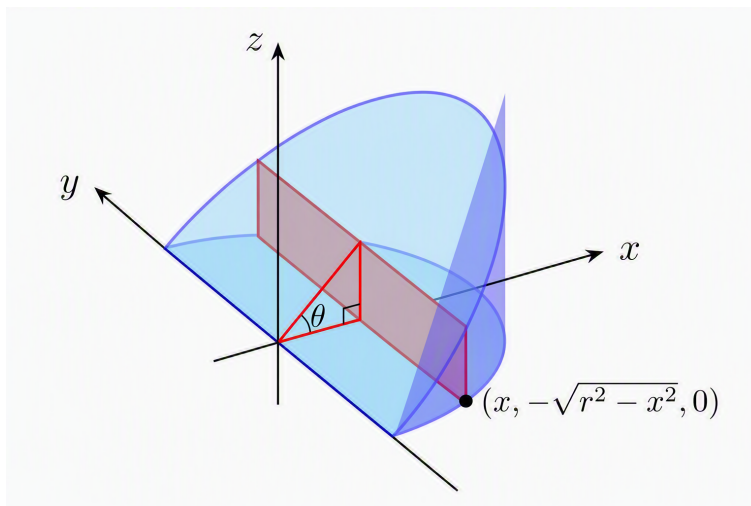


Lengths of the triangles sides: _____

Area of an equilateral triangle: _____

Set up the definite integral and calculate the volume:

Try it yourself. Find the volume of the wedge cut from a right circular cylinder of radius r by a plane passing through the y -axis and making an angle θ with the XY -plane, as shown in the figure.



Cross-sections perpendicular to the base are rectangles with dimensions:

length of side in the XY -plane: _____

length of other side: _____

Area of the rectangle: _____

Definite integral equal to the volume of the wedge: _____

Evaluating the integral:

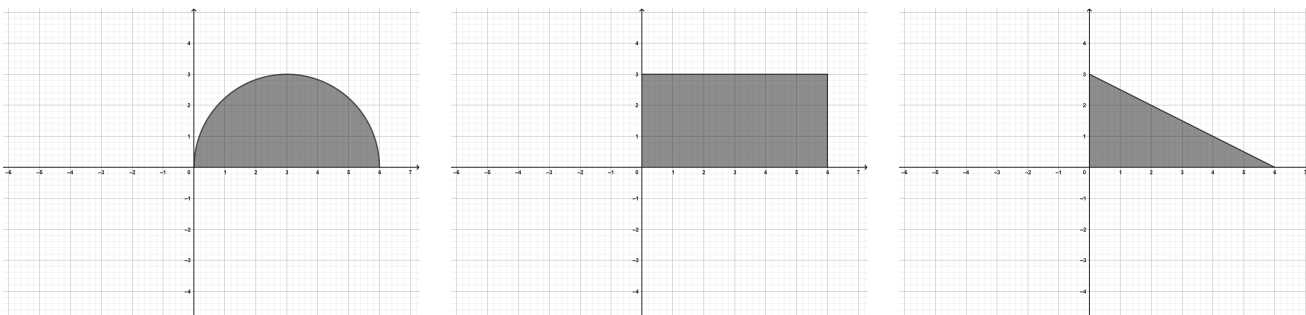
Definition 3.3 – Solid of revolution

A solid of revolution is the solid formed by revolving a plane region about one axis, called the axis of rotation of the solid.

In this course, we will only consider solids of revolution with an axis of rotation parallel to the x -axis or parallel to the y -axis.

Example 3.3

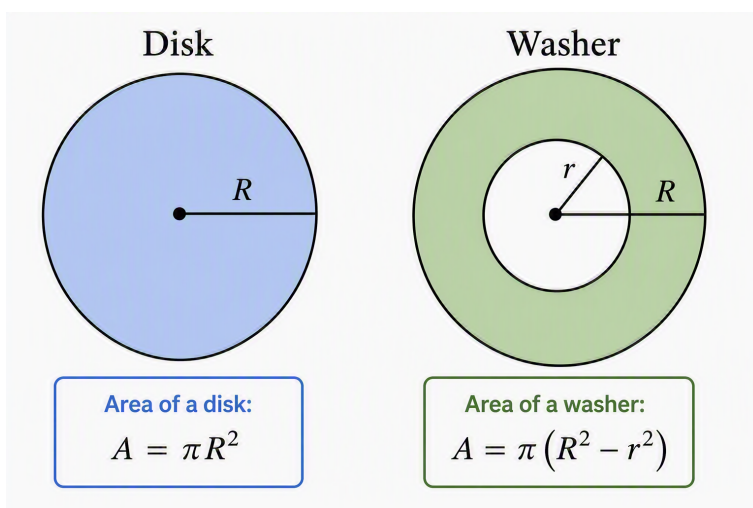
Which solid of revolution originates when each one of the following planar regions is rotated about the x -axis?



Cylinder

All solids of revolution share an important property: cross-sections perpendicular to the axis of rotation are either disks or washers.

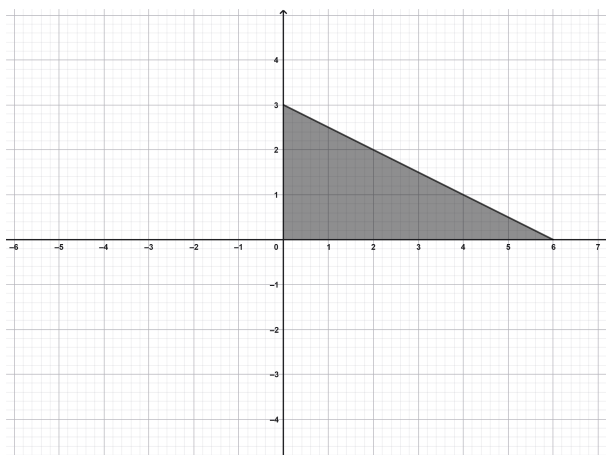
To find the volume of a solid of revolution, we use the slicing method and integrate the area of these cross-sections. Since the cross-sections are disks or washers, this technique is known as the disks/washers method.



Thus, to find the volume of a solid of revolution, we need only integrate a function of the form πR^2 or $\pi(R^2 - r^2)$. The key step is determining the radii R and r , as illustrated in the following examples.

Example 3.4

Let us find the volume of the cone that originates when the following region is rotated about the x -axis.



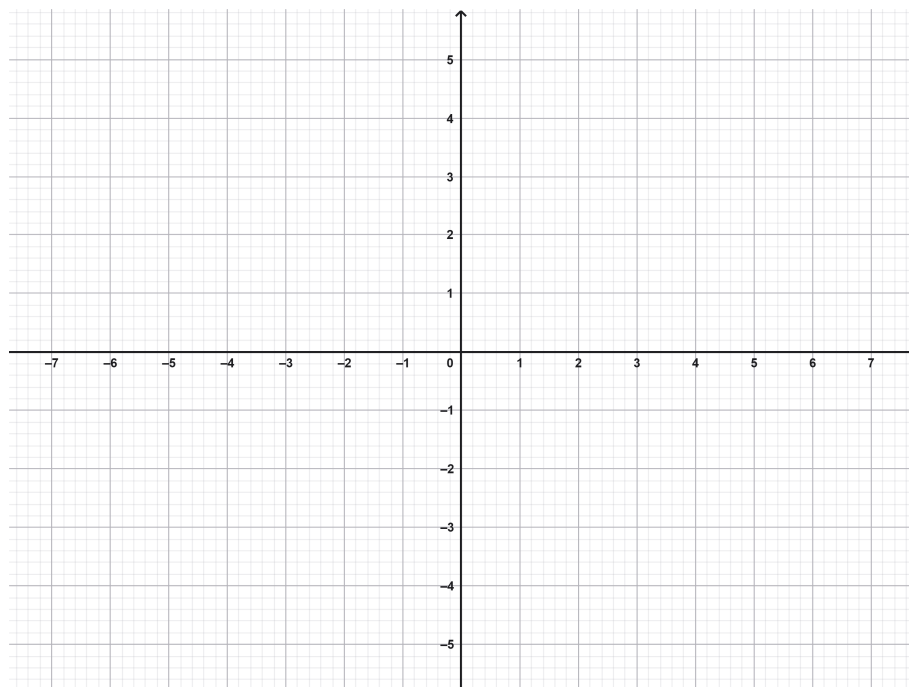
1. Sketch the region. Identify the axis of rotation.
2. Draw a representative slice (perpendicular to the axis of rotation).
3. Determine the variable and limits of integration: _____
4. Find the radius (as a function of the variable of integration): _____
5. Write and evaluate the integral(s):

Example 3.5

Let us find the volume of the solid obtained by rotating the region bounded by the graph of $f(x) = \sqrt{x}$, the x -axis, and the lines $x = 0$ and $x = 4$ about the x -axis.

This is a desmos activity that will help you understand the solid of revolution.

1. Sketch the region. Identify the axis of rotation.



2. Draw a representative slice.

3. Determine the variable and limits of integration: _____

4. Find radius or inner and outer radii (as functions of the variable of integration):

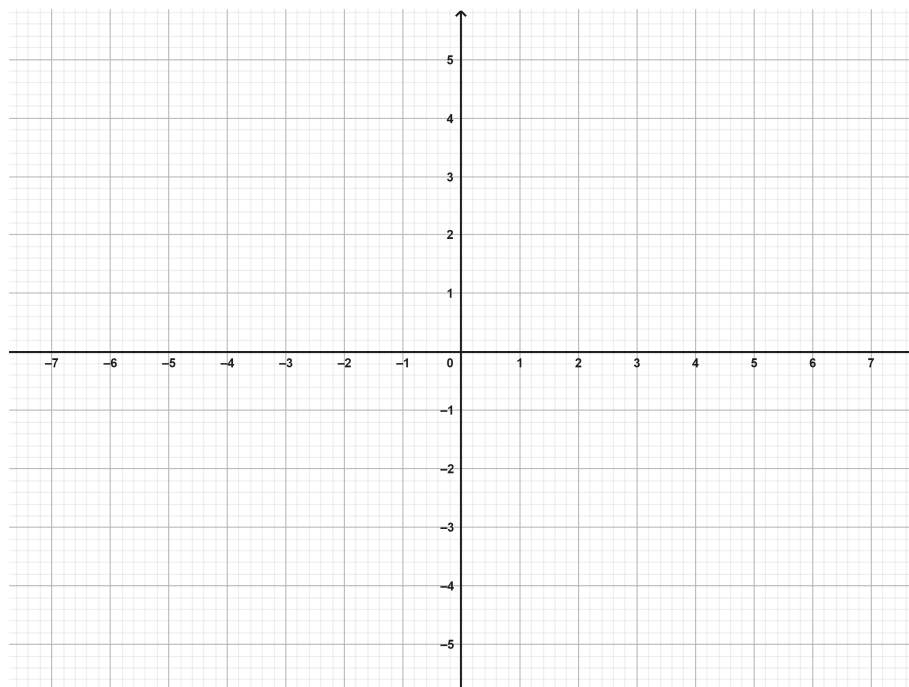
5. Write and evaluate the integral(s):

Example 3.6

Let us find the volume of the solid that originates when the same region above is rotated about the y -axis.

This is a desmos activity that will help you understand the solid of revolution.

1. Sketch the region. Identify the axis of rotation.



2. Draw a representative slice.

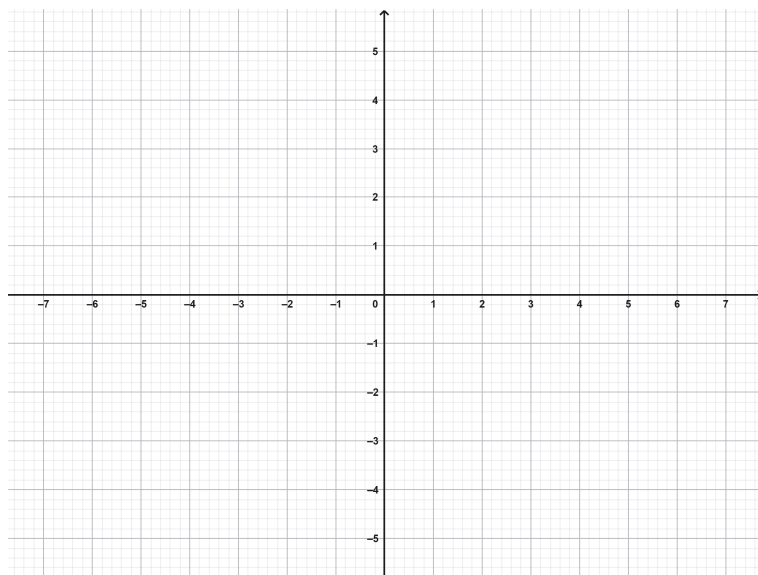
3. Determine the variable and limits of integration: _____

4. Find radius or inner and outer radii (as functions of the variable of integration): _____

5. Write and evaluate the integral(s):

Example 3.7

How does the integral we previously set up change if the axis of rotation is $x = -1$?

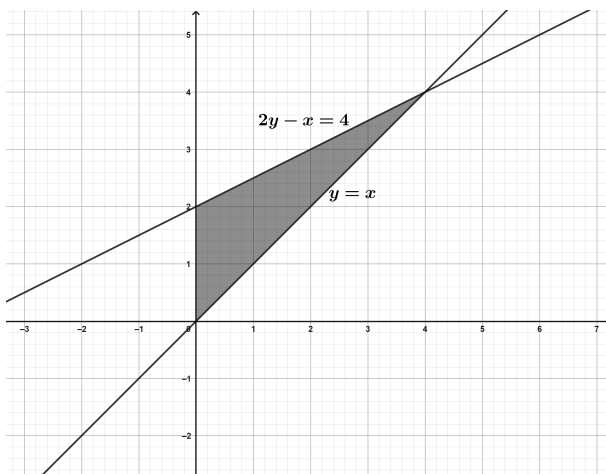


1. Sketch the region, including the axis of rotation.
2. Draw a representative slice.
3. Determine the variable and limits of integration.
4. Find the outer and inner radii.
5. Write and evaluate the integral(s).

The steps outlined in Table 1 are intended to help you set up the definite integral(s) that give the volume of a solid of revolution using the disks/washers method. These steps can be summarized as follows:

1. Sketch the region.
2. Draw a representative slice.
3. Determine the variable and limits of integration.
4. Find the outer and inner radii. The inner radius is zero when the cross-section is a disk.
5. Write and evaluate the integral(s).

Try it yourself. Determine the volume of the solid obtained by rotating the following region about the y -axis.



1. Sketch the region. Identify the axis of rotation.
2. Draw a representative slice.
3. Determine the variable and limits of integration: _____
4. Find radius or inner and outer radii (as functions of the variable of integration):

5. Write and evaluate the integral(s):

Now, let us practice these methods. Try to avoid these common mistakes.

! Common Mistakes

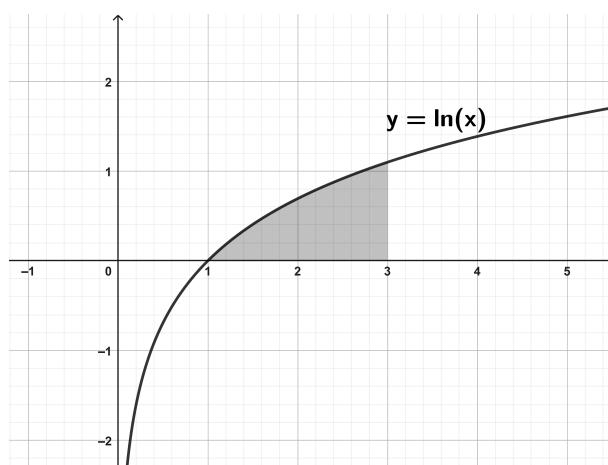
- Forgetting that the radii are always the *distance from the curve to the axis of rotation*, they are not always equal to the function values on the region boundary.
- Missing a gap between the region and the axis of rotation, and using the disk method (one radius) when a washer (inner *and* outer radius) is actually needed.

- Swapping inner and outer radius, so $(\text{inner radius})^2$ ends up larger than $(\text{outer radius})^2$ and the integral comes out negative.
- Forgetting to *square* the radius before multiplying by π , writing πR instead of πR^2 .
- Forgetting the formula for the area of a washer and writing it as $\pi(R - r)^2$, instead of $\pi(R^2 - r^2)$.

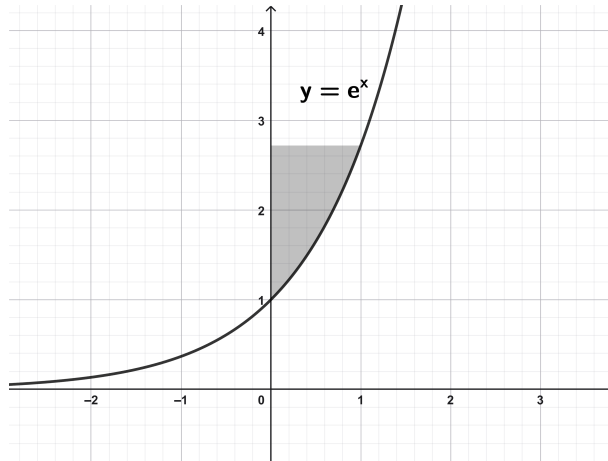
Practice Problems

Level I problems

1. Determine the volume of the solid generated by revolving the region bounded by the graphs of $f(x) = \sqrt{x} + 1$, $g(x) = x^2 + 1$ about the x -axis.
2. Let R be the region enclosed by $y = \ln(x)$, $y = 0$, and $x = 3$ (see figure below). Calculate the volume of the solid obtained by rotating R about the y -axis.



3. Let R be the region enclosed by $y = e^x$, and the y -axis when $0 \leq x \leq 1$ (see figure below). Calculate the volume of the solid obtained by rotating R about the x -axis.

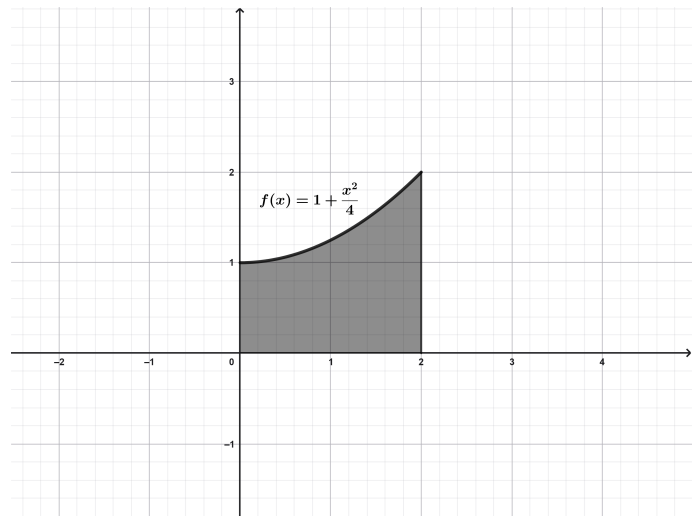


Practice Problems

Level II problems

- Find the volume of the following solid: the base is the region bounded from above by the graph of $f(x) = 2 - x^2$, $-1 \leq x \leq 1$ and from below by the graph of $g(x) = x^2$, $-1 \leq x \leq 1$. Parallel cross sections perpendicular to the base are circular disks.
- Find the volume of the solid generated by rotating the region bounded by the graphs of $x = \sqrt{y}$ and $x = \frac{y}{2}$ about the line $x = 4$.
- Calculate the volume of the solid of revolution generated by revolving the region in the figure below about the following axes of rotation.

- x -axis,
- y -axis,
- $y = 2$,
- $x = -1$,
- $x = 3$,
- $y = -1$.



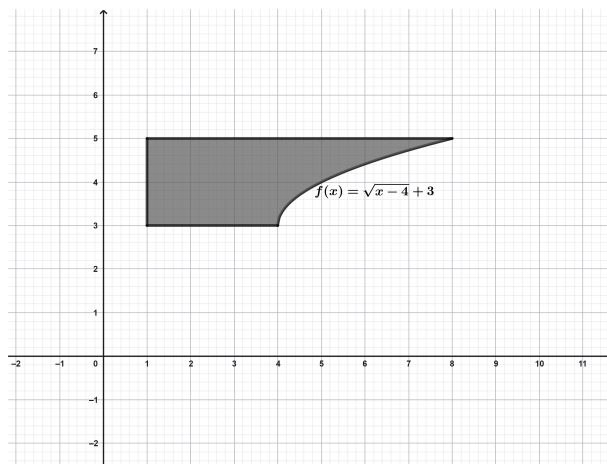
Practice Problems

Level III problems

1. Let $\mathcal{R}$ be the shaded region bounded by the lines

$$x = 1, y = 3, y = 5, \text{ and } f(x) = \sqrt{x-4} + 3$$

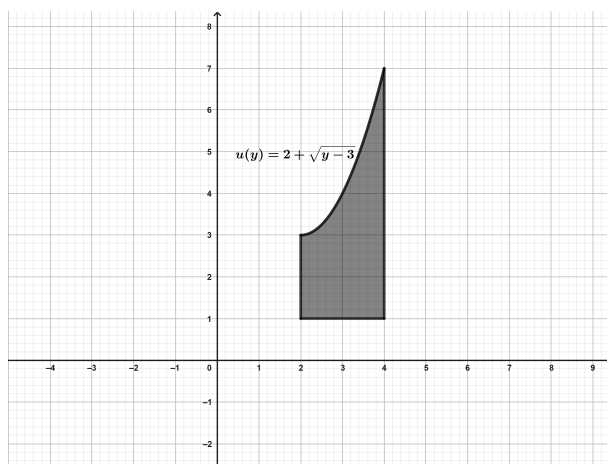
shown in the figure below. Set up the integral(s) to calculate the volume of the solid generated when $\mathcal{R}$ is rotated about the x -axis.



2. Let $\mathcal{R}$ be the shaded region bounded by the lines

$$x = 2, x = 4, y = 1, \text{ and } u(y) = 2 + \sqrt{y-3}$$

shown in the figure below. Set up the integral(s) equal to the volume of the solid generated when $\mathcal{R}$ is rotated about the y -axis.



Checking Your Understanding

1. Suppose $0 \leq f(x) \leq 5$ for all $x \in [a, b]$ and we want to calculate the volume of the solid that originates when the region bounded by the graph of $f(x)$ and the x -axis from $x = a$ to $x = b$ is rotated about the line $y = -1$. True or False: the definite integral to be evaluated if the disks/washers method is used is

$$\int_a^b \pi(f(x) - 1)^2 dx.$$

2. Suppose $0 \leq f(x) \leq 5$ for all $x \in [a, b]$. Set up the definite integral equal to the volume of the solid that originates when the region bounded by the graph of $f(x)$ and the x -axis from $x = a$ to $x = b$ is rotated about the line $y = 7$.
3. A classmate rotates the region between $y = \sqrt{x}$ and the x -axis on $[0, 4]$ about the line $y = -1$ (instead of about the x -axis), but still uses radius $= \sqrt{x}$ in the integral. What did they forget to account for, and what is the correct radius?
4. The washer method computes $\pi[(\text{outer radius})^2 - (\text{inner radius})^2]$, not $\pi(\text{outer} - \text{inner})^2$. Give a one-sentence geometric reason this has to be a *difference of squares*, not a squared difference.
5. Volume by slicing says $V = \int_a^b A(x) dx$ for a general cross-sectional area function $A(x)$. What is $A(x)$ specifically for the disk method? For the washer method? Explain why disks and washers are both special cases of this one general principle.

After this lesson I can:

- | | |
|---|--|
| ☐ Interpret volume as the accumulation of cross-sectional area. | ☐ Determine the radius, inner radius, and outer radius from a graph or labeled region. |
| ☐ Use the slicing method to set up one or definite integrals for the volume of a solid. | ☐ Explain how the axis of rotation affects the radius formulas. |
| ☐ Identify disk and washer cross-sections in a solid of revolution. | ☐ Construct and evaluate volume integrals using the disk/washer method. |

Muddiest Point: What concept from today's lesson is still unclear?

Step	What to do
Sketch the region and identify the axis of rotation	1. Draw the curves that enclose the region to be rotated. 2. Shade the region to be rotated. 3. Mark the axis of rotation. 4. Identify the coordinates of the leftmost, rightmost, lowest and highest points of the region. Let a and b be the x-coordinates of the leftmost and rightmost points. Let c and d be the y-coordinates of the lowest and highest points.
Draw a representative slice	When using the disks/washers method, draw slices that are perpendicular to the axis of rotation . Each slice intersects the boundary of the region at two points, N and F . Let N denote the point that is nearest to the axis of rotation, and let F denote the point that is farthest from the axis of rotation.
Determine the variable and limits of integration	1. Vertical slices: integrate with respect to x. The limits of integration are a and b (identified in step 1). 2. Horizontal slices: integrate with respect to y. The limits of integration are c and d (identified in step 1). Remember: Integrate with respect to the variable measured along the axis of rotation.
Find the function to be integrated	For each slice, identify the outer radius R and inner radius r. • The outer radius R is the distance from the axis of rotation to F (F was defined in step 2). • The inner radius r is the distance from the axis of rotation to N (N was defined in step 2). The area of a typical cross-section (represented by a slice) is $A = \pi (R^2 - r^2)$. Vertical slices: $A(x) = \pi (R(x)^2 - r(x)^2)$, $x \in [a, b]$. Horizontal slices: $A(y) = \pi (R(y)^2 - r(y)^2)$, $y \in [c, d]$. Remember: The radius is always the distance from the axis of rotation to a boundary curve. First identify the point on the slice that is farthest from the axis to determine R, and then identify the point that is nearest to the axis to determine r.
Write and evaluate the integral	Integrate the cross-sectional area over the appropriate interval. $V = \int_a^b A(x) \, dx \quad \text{or} \quad V = \int_c^d A(y) \, dy.$ Evaluate the integral to obtain the volume of the solid.

Table 1: Setting up the definite integral for the disks/washers method